

Dataset Description

Dataset Name : VinBigData Chest X-ray Abnormalities Detection

Overview:The VinBigData Chest X-ray Abnormalities Detection dataset is a large-scale medical imaging dataset used for detecting thoracic abnormalities in chest X-ray images. The dataset contains high-resolution chest X-rays along with bounding box annotations for multiple abnormality classes.

The dataset is designed for object detection tasks in the medical imaging domain.

Source:The dataset was obtained from the Kaggle competition: VinBigData Chest X-ray Abnormalities Detection.

Number of Samples:

- Total images in original dataset: ~18,000
- Dataset size: ~19.5 GB
- Subset used in this project: 1500 randomly selected images

Column Annotations:

The train.csv file contains:

- image_id – Unique identifier of each image
- class_id – Numerical abnormality label
- class_name – Abnormality name
- x_min, y_min, x_max, y_max – Bounding box coordinates
- rad_id – Radiologist ID

File Format:

- Images: .dicom (Digital Imaging and Communications in Medicine).
- Annotations: .csv
- Converted images (in this project): .jpg

Data Preprocessing Performed upto:

The following steps were completed in this project:

1. Loaded the dataset and examined the train.csv file.
2. Checked the structure and unique image IDs.
3. Randomly selected 1500 images for experimentation due to storage limitations.
4. Developed a script to convert selected .dicom images to .jpg format.
5. Verified the conversion by displaying sample converted images.